Bias Variance Tradeoff

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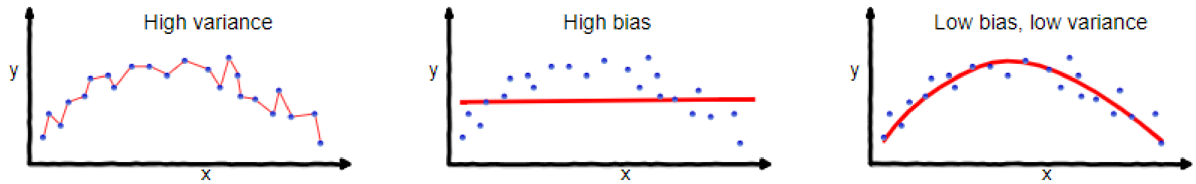
Evaluating the accuracy of a ML model is critical in selecting and deploying a machine learning model.

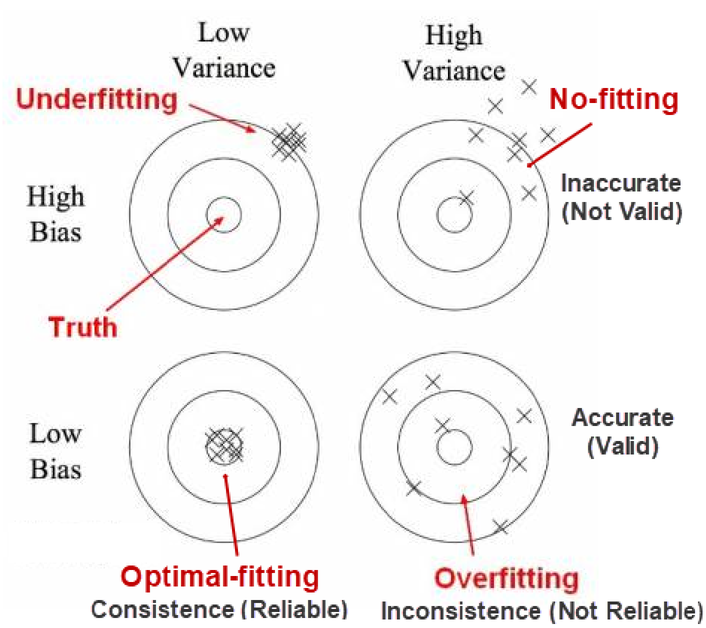


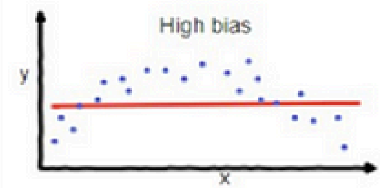
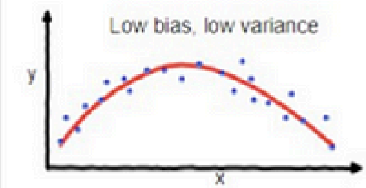
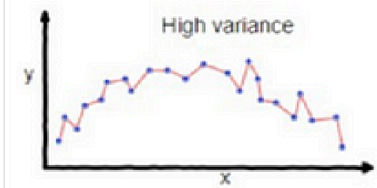
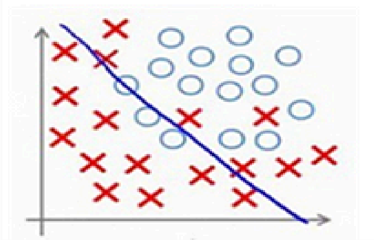
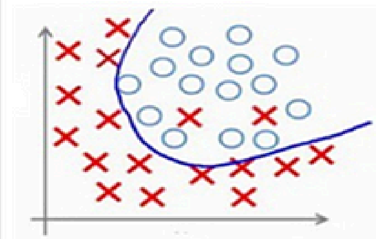
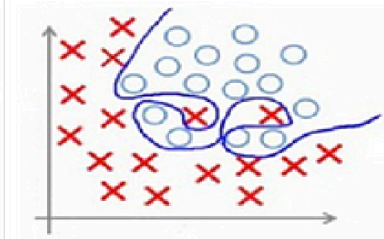
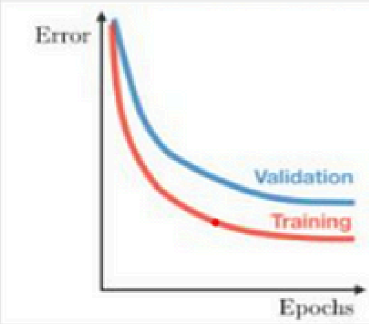
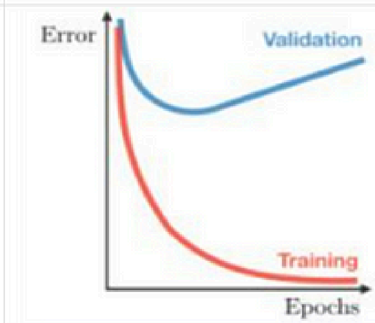
overfitting **underfitting** **Good balance**

Overfitting: *In model evaluating, overfitting occurs when a learning model customizes itself too much to describe the relationship between training data and the labels. Overfitting tends to make the model very complex by having too many parameters. By doing this, it loses its generalization power, which leads to poor performance on new [evaluation] data.*

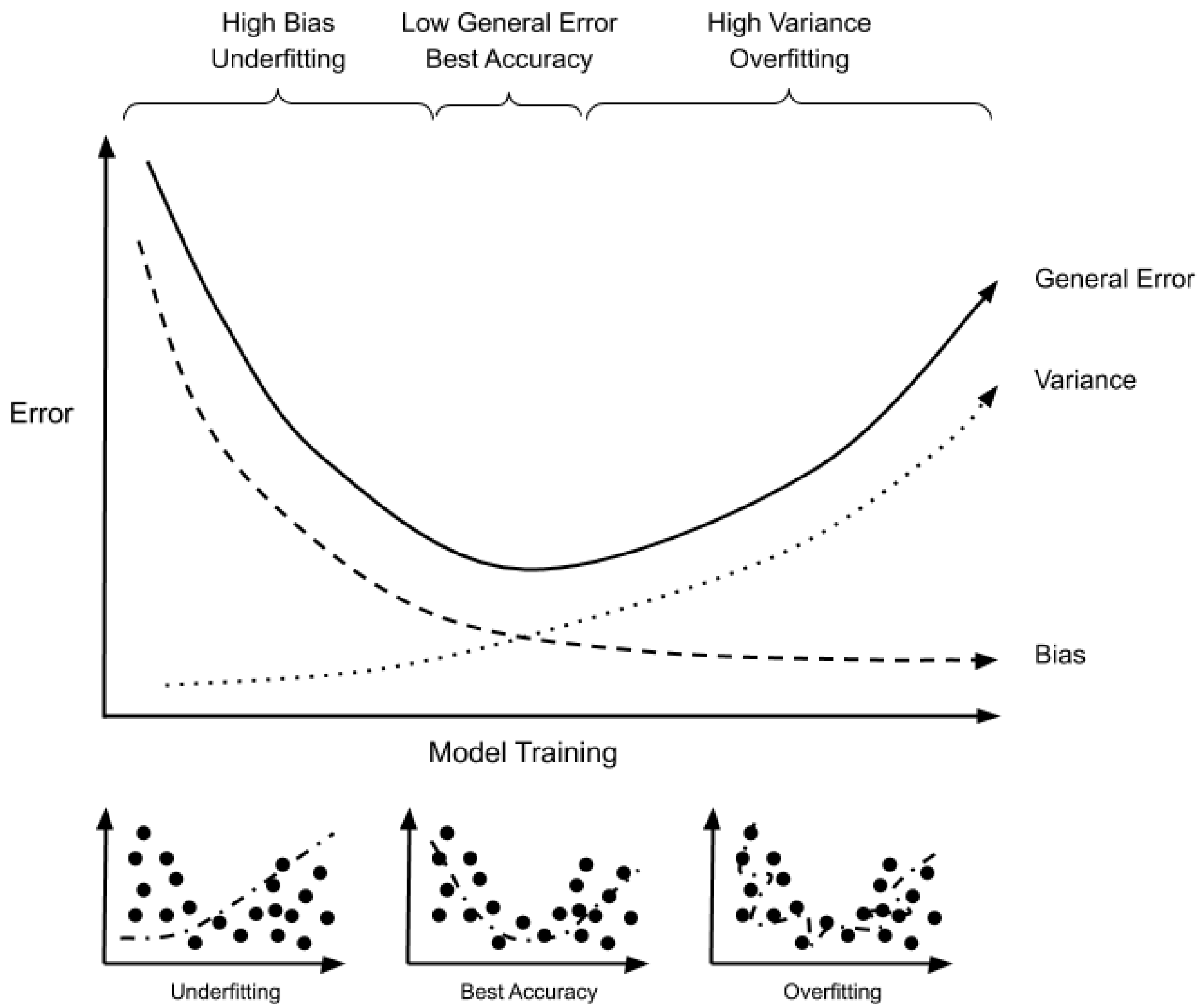
Underfitting: *In model evaluating,* a model is underfitting the training data when the model performs poorly on the training data. This is because the model is unable to capture the relationship between the input examples (often called X) and the target values (often called y).

Model fit depends on solving the issue and balancing the tradeoff between bias and variance.



	Underfitting	Optimal-fitting	Overfitting
Symptoms	• High Bias • High training error • Training error close to Testing error • Unable to capture the relationship between training data and labels • Inaccurate (not valid)	• Low bias, low variance • Optimize training error • Training error slightly lower than test error • Capture well the relationship between training data and labels • Accurate (valid) and Consistence (reliable)	• High Variance • Very low training error • Training error much lower than test error • Customize too much the relationship between training data and labels • Inconsistence (not reliable)
Regression			
Classification			
Deep Learning			
Remedies	• Increase training data • Reduce noise • Perform feature scaling (Normalization or Standardization) • Train longer • Add more features • Increase complexity • Combine models using Boosting ensemble • Increase capacity • Change network structure or parameters • Add inputs, nodes, layers		• Use data augmentation • Reduce noise • Perform feature scaling • Perform regularization • Use cross-validation • Early stopping • Reduce features • Simplify complexity • Combine models using Bagging ensemble • Pruning number of weights (structure) and values (parameters) • Dropout inputs or units

Overfitting: High Variance, Low Bias ***Underfitting: High Bias, Low Variance***



References

[1] https://github.com/visualmatics/Accuracy-The-Bias-Variance-Tradeoff